

Learning Management System Analytics Pipeline

1. Abstract

LearnTrack is a batch data engineering pipeline designed to transform raw Learning Management System (LMS) records into clean, enriched, and business-ready analytical datasets. The project follows the Medallion Architecture using Bronze, Silver, and Gold layers.

The analytical layer covers course completion, learner engagement, instructor effectiveness, assessment performance, dropout behaviour, re-enrolment patterns, category performance, learner progress, and overall LMS KPIs.

2. Introduction

Learning platforms continuously generate operational data whenever learners enrol in courses, progress through content, complete assessments, receive certificates, or discontinue learning. Raw operational data is not directly suitable for reliable business analysis.

LearnTrack builds the data engineering layer required to structure, clean, enrich, and aggregate this information so that platform administrators, instructional designers, and management teams can use consistent analytical outputs.

3. Problem Statement

Learning management platforms generate rich learner interaction data every day, but without a structured analytics pipeline it is difficult to answer questions about learner engagement, course completion, instructor effectiveness, assessment performance, dropout, and re-enrolment.

4. Objectives

- Build a reliable batch data engineering pipeline for LMS data.
- Preserve raw source data in the Bronze layer.
- Establish data quality and enrichment in the Silver layer.
- Create business-oriented analytical datasets in the Gold layer.
- Analyze course completion and dropout behaviour.
- Analyze learner engagement and progress.
- Evaluate instructor effectiveness and assessment performance.
- Identify re-enrolment patterns and overall platform KPIs.

5. Dataset Description

The project proposal defines three source datasets covering a 90-day operational window from January to March 2024.

Dataset	Records
learners.csv	500
courses.csv	60
enrolment_activity.csv	2,000

Intentional data-quality issues include 10 duplicate enrolment rows, blank instructor names, blank last-activity dates, NULL completion dates for non-completed enrolments, and NULL assessment scores for non-completed enrolments.

6. Technology Stack

- Python – data generation and ingestion scripting.
- Apache PySpark – distributed data processing and transformations.
- Databricks – managed Spark runtime and notebook development.
- Delta Lake – storage layer across pipeline tiers.
- SQL – analytical querying and business-facing access to Gold outputs.

7. System Architecture

7.1 Bronze Layer – Raw Ingestion

Preserves source data with minimal modification, including source-level NULLs, duplicates, and inconsistencies.

```
print("Bronze Learners:")
spark.read.format("delta").load("/content/bronze/learners").show(3)

print("Bronze Courses:")
spark.read.format("delta").load("/content/bronze/courses").show(3)

print("Bronze Enrolment Activity:")
spark.read.format("delta").load("/content/bronze/enrolment_activity").show(3)
```

```
*** Bronze Learners:
```

learner_id	learner_name	email	phone_number	city	registration_date	subscription_type
LN0001	Ananya Sharma	ananya.sharma@abc...	9833218196	Mumbai	2022-04-06	Free
LN0002	Sachin Pillai	sachin.pillai@xyz...	9083863794	Indore	2022-06-13	Premium
LN0003	Suresh Mishra	suresh.mishra@ghi...	9235116155	Chennai	2022-02-14	Premium

only showing top 3 rows

```
Bronze Courses:
```

course_id	course_title	category	instructor_id	instructor_name	duration_hours	difficulty_level	price_inr
CRS001	Python for Data S...	Data Science	INS009	Suresh Gupta	15	Intermediate	2986
CRS02	Machine Learning ...	AI & ML	INS010	Priya Singh	18	Beginner	223
CRS03	Deep Learning wit...	AI & ML	INS004	Sunita Rao	19	Advanced	13612

only showing top 3 rows

```
Bronze Enrolment Activity:
```

enrolment_id	learner_id	course_id	enrol_date	expected_completion_date	actual_completion_date	status	progress_pct	last_activity_date	assessment_score	attempts	feedback_rating	certificate_issued
ENR00460	LN00376	CRS001	2024-01-13	2024-02-12	2024-02-21	Completed	100	2024-02-21	79.22	1	2	Yes
ENR00681	LN0486	CRS055	2024-01-16	2024-01-26	2024-01-19	Completed	100	2024-01-19	56.62	1	3	No
ENR0477	LN00352	CRS044	2024-03-16	2024-06-04	2024-03-11	Completed	100	2024-03-11	93.98	1	4	Yes

only showing top 3 rows

7.2 Silver Layer – Cleaning and Enrichment

Standardizes dates and types, removes duplicate enrolments, resolves instructor names using identifiers, derives learning duration, and joins learner, course, and enrolment context.

```
silver_table = (
    spark.read
    .format("delta")
    .load("/content/silver/enriched_enrolments")
)

print("Silver Records:", silver_table.count())

silver_table.show(5, truncate=False)
```

... Silver Records: 1998

course_id	learner_id	enrolment_id	enrol_date	expected_completion_date	actual_completion_date	status	progress_pct	last_activity_date	assessment_score	attempts	feedback_rating	certificate_issued	learning_duration	learner
CRS045	LRN0467	ENR00001	2024-01-09	2024-01-19	NULL	Dropped	21	NULL	NULL	1	NULL	No	NULL	Ananya
CRS056	LRN0337	ENR00002	2024-02-24	2024-04-24	NULL	In Progress	44	NULL	NULL	1	NULL	No	NULL	Ananya
CRS025	LRN0163	ENR00003	2024-03-14	2024-04-23	2024-03-31	Completed	100	2024-03-31	50.37	1	2	No	17	Shalini
CRS052	LRN0258	ENR00004	2024-02-13	2024-05-03	2024-03-17	Completed	100	2024-03-17	64.98	1	4	Yes	33	Sunita
CRS051	LRN0499	ENR00005	2024-01-10	2024-03-30	NULL	Dropped	13	2024-01-18	NULL	1	5	No	NULL	Divya P

only showing top 5 rows

7.3 Gold Layer – Business Metrics

Contains business-oriented analytical datasets for course completion, instructor performance, learner engagement, assessment performance, re-enrolment, dropout, category performance, learner progress, and platform KPIs.

8. Data Processing Pipeline

8.1 Ingestion

Load the three CSV source datasets into the Bronze layer.

8.2 Data Quality

Identify and handle duplicates, missing values, inconsistent fields, and required type conversions.

8.3 Enrichment

Resolve instructor information and combine learner, course, and enrolment context.

8.4 Derived Metrics

Calculate learning duration and analytical measures required by downstream reporting.

8.5 Aggregation

Generate course, learner, instructor, category, assessment, dropout, re-enrolment, and KPI datasets.

8.6 Gold Storage and Validation

Persist analytical outputs in Delta format and validate the generated Gold datasets.

9. Gold Layer Analytics

9.1 Course Completion

Measures total enrolments, completed enrolments, and completion rates by course and classifies completion performance.

```
gold_course_completion = (  
    gold_course_completion  
    .withColumn(  
        "completion_category",  
        when(col("completion_rate") >= 70, "High")  
        .when(col("completion_rate") >= 50, "Moderate")  
        .otherwise("At Risk")  
    )  
)  
  
gold_course_completion.select(  
    "course_id",  
    "course_title",  
    "completion_rate",  
    "completion_category"  
)>.show(10, truncate=False)
```

```
+-----+-----+-----+-----+  
|course_id|course_title|completion_rate|completion_category|  
+-----+-----+-----+-----+  
|CRS018|Flutter App Development|34.29|At Risk|  
|CRS024|Computer Vision|45.0|At Risk|  
|CRS011|Figma for Beginners|56.52|Moderate|  
|CRS036|Advanced Python OOP|31.82|At Risk|  
|CRS053|Supply Chain Management|53.33|Moderate|  
|CRS026|Full Stack Django|28.21|At Risk|  
|CRS021|Power BI & Tableau|38.24|At Risk|  
|CRS007|Business Analytics Essentials|35.48|At Risk|  
|CRS017|Network Security Essentials|32.43|At Risk|  
|CRS043|Leadership & Communication|34.29|At Risk|  
+-----+-----+-----+-----+  
only showing top 10 rows
```

```
gold_instructor_performance = (  
    silver_table  
    .groupBy(  
        "instructor_id",  
        "instructor_name"  
    )  
    .agg(  
        count("*").alias("total_enrolments"),  
        count(  
            when(col("status") == "Completed", True)  
        ).alias("completed_enrolments")  
    )  
    .withColumn(  
        "completion_rate",  
        round(  
            col("completed_enrolments") /  
            col("total_enrolments") * 100,  
            2  
        )  
    )  
    .orderBy(col("completion_rate").desc())  
)  
  
gold_instructor_performance.show(10, truncate=False)
```

```
+-----+-----+-----+-----+-----+  
|instructor_id|instructor_name|total_enrolments|completed_enrolments|completion_rate|  
+-----+-----+-----+-----+-----+  
|INS009|Suresh Gupta|170|83|48.82|  
|INS002|Meera Nair|159|75|47.17|  
|INS010|Anjali Iyer|176|77|43.75|  
|INS001|Rajiv Sharma|60|26|43.33|  
|INS007|Vikas Mehta|141|61|43.26|  
|INS014|Pooja Mishra|65|28|43.08|  
|INS005|Deepak Joshi|103|44|42.72|  
|INS012|Nisha Agarwal|51|21|41.18|  
|INS006|Priya Singh|250|100|40.0|  
|INS004|Sunita Rao|203|80|39.41|  
+-----+-----+-----+-----+-----+
```

9.2 Instructor Performance

Combines enrolment volume, completed enrolments, completion rate, average assessment score, and assessment attempts.

```
gold_course_completion.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/gold/course_completion")

gold_instructor_performance.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/gold/instructor_performance")

print("Course Completion and Instructor Performance Gold tables saved successfully!")
```

Course Completion and Instructor Performance Gold tables saved successfully!

9.3 Learner Engagement

Uses latest activity, progress percentage, and status to classify learners into engagement categories.

```
gold_learner_engagement = (
    learner_latest_activity
    .withColumn(
        "engagement_category",
        when(col("status") == "Dropped", "Dropped")
        .when(col("status") == "Completed", "Completed")
        .when(
            (col("status") == "In Progress") &
            (col("progress_pct") < 50),
            "Low Engagement"
        )
        .otherwise("Active")
    )
)
```

```
gold_learner_engagement.show(10, truncate=False)
```

learner_id	learner_name	last_activity_date	progress_pct	status	engagement_category
LRN0001	Ananya Sharma	2024-03-24	90	In Progress	Active
LRN0002	Sachin Pillai	2024-03-02	24	Dropped	Dropped
LRN0003	Suresh Mishra	2024-03-03	15	Dropped	Dropped
LRN0004	Vijay Kumar	2024-03-28	100	Completed	Completed
LRN0005	Priya Chatterjee	2024-03-01	77	In Progress	Active
LRN0006	Karan Pillai	2024-03-20	13	In Progress	Low Engagement
LRN0007	Aditya Bose	2024-03-23	100	Completed	Completed
LRN0008	Divya Yadav	2024-03-12	0	Dropped	Dropped
LRN0009	Rohan Chatterjee	2024-02-16	71	In Progress	Active
LRN0010	Aarav Desai	2024-03-31	17	In Progress	Low Engagement

only showing top 10 rows

```
gold_learner_engagement.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/gold/learner_engagement")

print("Learner Engagement Gold table saved successfully!")
```

Learner Engagement Gold table saved successfully!

9.4 Assessment Performance

Analyzes average assessment score, total attempts, and assessed learners.

```
gold_assessment_performance = (
    gold_assessment_performance
    .withColumn(
        "assessment_category",
        when(col("average_score") >= 70, "Good")
        .when(col("average_score") >= 50, "Needs Attention")
        .otherwise("Bottleneck")
    )
)

gold_assessment_performance.select(
    "course_id",
    "course_title",
    "average_score",
    "total_attempts",
    "assessment_category"
).show(10, truncate=False)
```

course_id	course_title	average_score	total_attempts	assessment_category
CRS058	Zero Trust Security	81.61	15	Good
CRS004	React.js Complete Guide	79.76	21	Good
CRS006	HTML & CSS Mastery	77.98	15	Good
CRS028	Entrepreneurship 101	77.55	20	Good
CRS022	Statistics for Data Science	76.31	11	Good
CRS011	Figma for Beginners	75.12	17	Good
CRS036	Advanced Python OOP	74.95	12	Good
CRS055	Motion UI Design	74.72	27	Good
CRS032	DevOps CI/CD Pipeline	74.47	13	Good
CRS001	Python for Data Science	74.33	23	Good

only showing top 10 rows

```
gold_assessment_performance.groupBy(
  "assessment_category"
).count().orderBy(
  col("count").desc()
).show()
```

```
+-----+-----+
|assessment_category|count|
+-----+-----+
|          Good|    35|
| Needs Attention|    25|
+-----+-----+
```

9.5 Re-enrolment

Identifies repeated learner-course enrolments and assessment attempts.

```
gold_reenrolment = (
  silver_table
  .groupBy(
    "learner_id",
    "learner_name",
    "course_id",
    "course_title"
  )
  .agg(
    count("*").alias("enrolment_count"),
    spark_max("attempts").alias("max_attempts")
  )
  .filter(
    (col("enrolment_count") > 1) |
    (col("max_attempts") >= 2)
  )
  .orderBy(
    col("enrolment_count").desc()
  )
)
```

```
gold_reenrolment.show(10, truncate=False)
```

```
*** +-----+-----+-----+-----+-----+-----+
|learner_id|learner_name |course_id|course_title          |enrolment_count|max_attempts|
+-----+-----+-----+-----+-----+-----+
|LRN0298   |Sunita Agarwal|CRS046   |Terraform Infrastructure|2              |2          |
|LRN0133   |Ravi Jain     |CRS055   |Motion UI Design       |2              |2          |
|LRN0275   |Manish Reddy  |CRS002   |Machine Learning Fundamentals|2              |2          |
|LRN0352   |Ajay Bose     |CRS044   |Adobe Illustrator Mastery|2              |2          |
|LRN0087   |Ishaan Shah   |CRS051   |MLOps & Model Deployment |2              |1          |
|LRN0036   |Megha Iyer    |CRS018   |Flutter App Development |2              |1          |
|LRN0046   |Pooja Patel   |CRS013   |AWS Solutions Architect |2              |1          |
|LRN0100   |Ritika Mehta  |CRS054   |Product Management     |2              |1          |
|LRN0430   |Reena Reddy   |CRS029   |Brand Identity Design   |2              |2          |
|LRN0490   |Rahul Reddy   |CRS054   |Product Management     |2              |3          |
+-----+-----+-----+-----+-----+-----+
```

only showing top 10 rows

```
print(
    "Total Re-enrolment Candidates:",
    gold_reenrolment.count()
)
```

Total Re-enrolment Candidates: 460

9.6 Dropout Analysis

Calculates course and platform-level dropout metrics.

```
gold_reenrolment.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/gold/reenrolment_analysis")

print("Re-enrolment Gold table saved successfully!")
```

Re-enrolment Gold table saved successfully!

```
total_enrolments = silver_table.count()

total_dropped = (
    silver_table
    .filter(col("status") == "Dropped")
    .count()
)

overall_dropout_rate = (
    total_dropped / total_enrolments
) * 100

overall_dropout_rate = float(
    f"{overall_dropout_rate:.2f}"
)

print("Total Enrolments:", total_enrolments)
print("Total Dropped:", total_dropped)
print("Overall Dropout Rate:", overall_dropout_rate, "%")
```

Total Enrolments: 1990
Total Dropped: 411
Overall Dropout Rate: 20.65 %

```
gold_dropout.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/content/gold/dropout_analysis")

print("Dropout Gold table saved successfully!")
```

Dropout Gold table saved successfully!

9.7 Category Performance

Compares categories using enrolments, completions, dropouts, completion rate, and dropout rate.

```
gold_category_performance = (  
  silver_table  
    .groupBy("category")  
    .agg(  
      count(".*").alias("total_enrolments"),  
      count(  
        when(col("status") == "Completed", True)  
      ).alias("completed_enrolments"),  
      count(  
        when(col("status") == "Dropped", True)  
      ).alias("dropped_enrolments")  
    )  
    .withColumn(  
      "completion_rate",  
      round(  
        col("completed_enrolments") /  
        col("total_enrolments") * 100,  
        2  
      )  
    )  
    .withColumn(  
      "dropout_rate",  
      round(  
        col("dropped_enrolments") /  
        col("total_enrolments") * 100,  
        2  
      )  
    )  
    .orderBy(  
      col("completion_rate").desc()  
    )  
)  
  
gold_category_performance.show(  
  truncate=False  
)
```

```
*** +-----+-----+-----+-----+-----+  
|category|total_enrolments|completed_enrolments|dropped_enrolments|completion_rate|dropout_rate|  
+-----+-----+-----+-----+-----+  
|Mobile Development|184|91|29|49.46|15.76|  
|AI & ML|253|114|51|45.06|20.16|  
|Design|249|109|55|43.78|22.09|  
|Data Science|291|117|68|40.21|23.37|  
|Business|300|119|63|39.67|21.0|  
|Web Development|290|115|62|39.66|21.38|  
|Cloud Computing|231|90|43|38.96|18.61|  
|Cybersecurity|192|63|40|32.81|20.83|  
+-----+-----+-----+-----+-----+
```

```
gold_category_performance.write \  
  .format("delta") \  
  .mode("overwrite") \  
  .save("/content/gold/category_performance")
```

```
print("Category Performance Gold table saved successfully!")
```

```
Category Performance Gold table saved successfully!
```

9.8 Learner Progress

Segments learners using latest progress percentage.

```
gold_progress_distribution = (  
  gold_learner_engagement  
  .withColumn(  
    "progress_category",  
    when(col("progress_pct") < 25, "Very Low")  
    .when(col("progress_pct") < 50, "Low")  
    .when(col("progress_pct") < 75, "Moderate")  
    .otherwise("High")  
  )  
)  
  
gold_progress_distribution.groupBy(  
  "progress_category"  
) .count() .orderBy(  
  col("count").desc()  
) .show()
```

```
+-----+-----+  
|progress_category|count|  
+-----+-----+  
|           High|  290|  
|       Very Low|   98|  
|           Low|   64|  
|       Moderate|   43|  
+-----+-----+
```

```
gold_progress_distribution.write \  
  .format("delta") \  
  .mode("overwrite") \  
  .save("/content/gold/progress_distribution")  
  
print("Progress Distribution Gold table saved successfully!")
```

```
Progress Distribution Gold table saved successfully!
```

9.9 Platform KPIs

Provides overall learner, course, enrolment, completion, and dropout KPIs.

```
id="kpi2"
total_learners = silver_table.select("learner_id").distinct().count()
total_courses = silver_table.select("course_id").distinct().count()
total_enrolments = silver_table.count()

total_completed = (
    silver_table
    .filter(col("status") == "Completed")
    .count()
)

total_dropped = (
    silver_table
    .filter(col("status") == "Dropped")
    .count()
)

overall_completion_rate = float(
    f"{{(total_completed / total_enrolments) * 100:.2f}}"
)

overall_dropout_rate = float(
    f"{{(total_dropped / total_enrolments) * 100:.2f}}"
)

print("==== LMS GOLD KPI SUMMARY ====")
print("Total Learners:", total_learners)
print("Total Courses:", total_courses)
print("Total Enrolments:", total_enrolments)
print("Completed Enrolments:", total_completed)
print("Dropped Enrolments:", total_dropped)
print("Overall Completion Rate:", overall_completion_rate, "%")
print("Overall Dropout Rate:", overall_dropout_rate, "%")
```

```
==== LMS GOLD KPI SUMMARY ====
Total Learners: 495
Total Courses: 60
Total Enrolments: 1990
Completed Enrolments: 818
Dropped Enrolments: 411
Overall Completion Rate: 41.11 %
Overall Dropout Rate: 20.65 %
```

10. Results and Key Findings

The completed pipeline produced the following overall platform metrics:

Metric	Value
Total learners	495
Total courses	60
Total enrolments	1,990
Completed enrolments	818
Dropped enrolments	411
Overall completion rate	41.11%
Overall dropout rate	20.65%

11. Business Insights

- Course completion and dropout rates identify courses requiring review.
- Instructor performance can be assessed using learner outcomes and assessment performance.
- Learner engagement categories help identify active, low-engagement, completed, and dropped learners.
- Assessment metrics highlight stronger and weaker course outcomes.
- Category-level comparison supports content planning and prioritisation.
- Platform KPIs provide a concise view of overall LMS performance.

12. Validation and Data Quality

The Silver layer handles duplicate enrolments, date and type standardization, instructor-name resolution, and learning-duration derivation. Gold outputs are read back from Delta storage and validated to ensure successful creation and expected analytical records.

13. Challenges and Solutions

Duplicate enrolment records: Deduplicate using the enrolment identifier before downstream aggregation.

Missing instructor names: Resolve names using the instructor identifier.

NULL completion and assessment fields: Retain meaningful NULLs for non-completed activity and apply metric-specific analytical logic.

Multiple source tables: Join learner, course, and enrolment activity in Silver.

Business-ready outputs: Materialize focused Gold aggregations and classifications.

14. Conclusion

LearnTrack demonstrates an end-to-end batch data engineering workflow for LMS analytics using the Medallion Architecture. The pipeline preserves raw source information, establishes data quality and business context in Silver, and produces business-ready analytical datasets in Gold.

The resulting outputs provide a foundation for monitoring LMS performance, understanding learner behaviour, evaluating courses and instructors, and supporting data-driven operational decisions.

15. Future Scope

- Integrate with a live operational LMS source.
- Schedule recurring pipeline execution.
- Add automated monitoring and alerts for learner inactivity and dropout risk.
- Expose Gold datasets through dashboards and BI tools.
- Extend historical tracking for changing instructor attributes.
- Scale processing patterns to larger LMS datasets.

16. References

LearnTrack: LMS Analytics Pipeline – Project Proposal & Technical Overview, Data Engineering Programme, Celebal Technologies.

Project implementation notebook: `LearnTrack_LMS_Analytics.ipynb`.